Bayesian inference in non-Gaussian time series with global-local shrinkage process priors

Mattias Villani

Department of Statistics, Stockholm University

Collaborators

- **Robert Kohn**, University of New South Wales, Sydney.
- **Ganna Fagerberg**, Stockholm University
- **Yijie Niu**, Stockholm University
- **Oskar Gustafsson**, Stockholm University

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Slides

mattiasvillani.com/seminars/nordstat2026

Julia packages (unoptimized, ongoing)

https://github.com/compbayes

Outline

0. Time-varying parameters and state-space models
1. Dynamic shrinkage process priors
2. Time-varying multi-seasonal ARMA
3. Non-Gaussian models
4. Posterior inference

Time-varying parameter models

- Regression with **time-varying parameters**

$$\begin{aligned} y_t &= \boldsymbol{x}_t^\top \boldsymbol{\beta}_t + \varepsilon_t & \varepsilon_t &\overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \\ \boldsymbol{\beta}_t &= \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_\nu) \end{aligned}$$

- **Linear Gaussian state-space model**

$$\begin{aligned} &\text{Observation model:} & \boldsymbol{y}_t &= \boldsymbol{C}_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ &\text{Transition model:} & \boldsymbol{\theta}_t &= \boldsymbol{A}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

with state $\boldsymbol{\theta}_t = \boldsymbol{\beta}_t$, $\boldsymbol{C}_t = \boldsymbol{x}_t^\top$ and $\boldsymbol{A}_t = \boldsymbol{I}_p$.

Regression with time-varying parameters

time series length

100

σ_ε

1

σ_{β_0}

1

σ_{β_1}

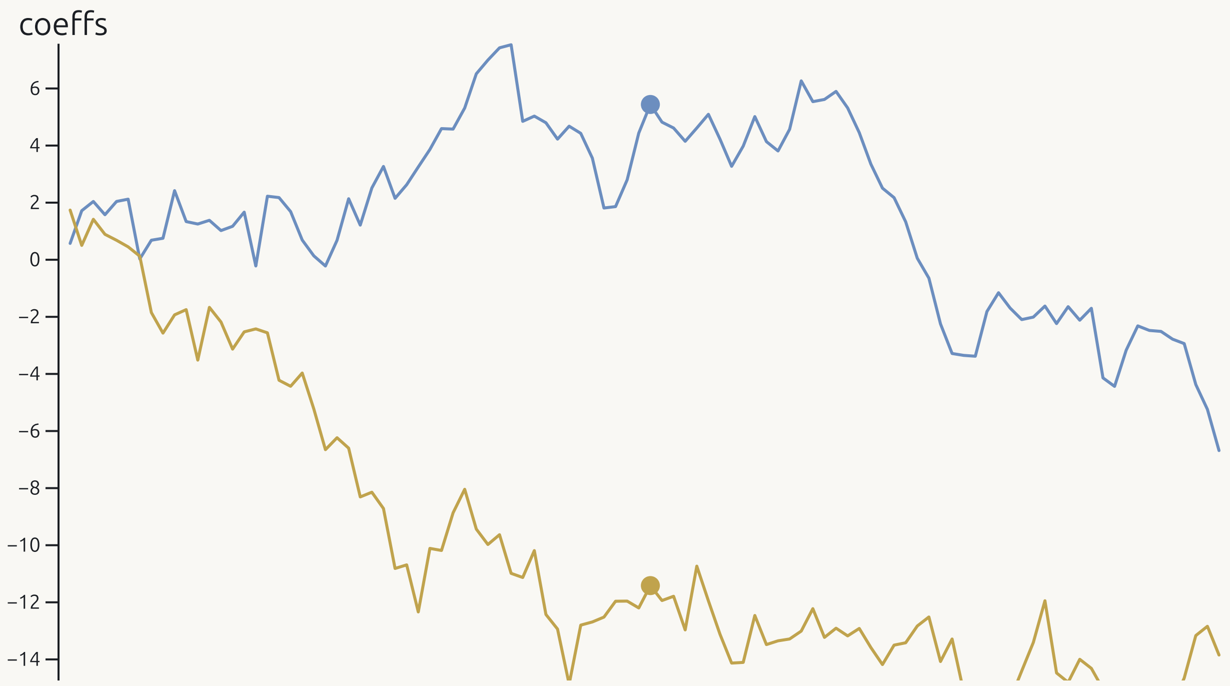
1

regr line at time

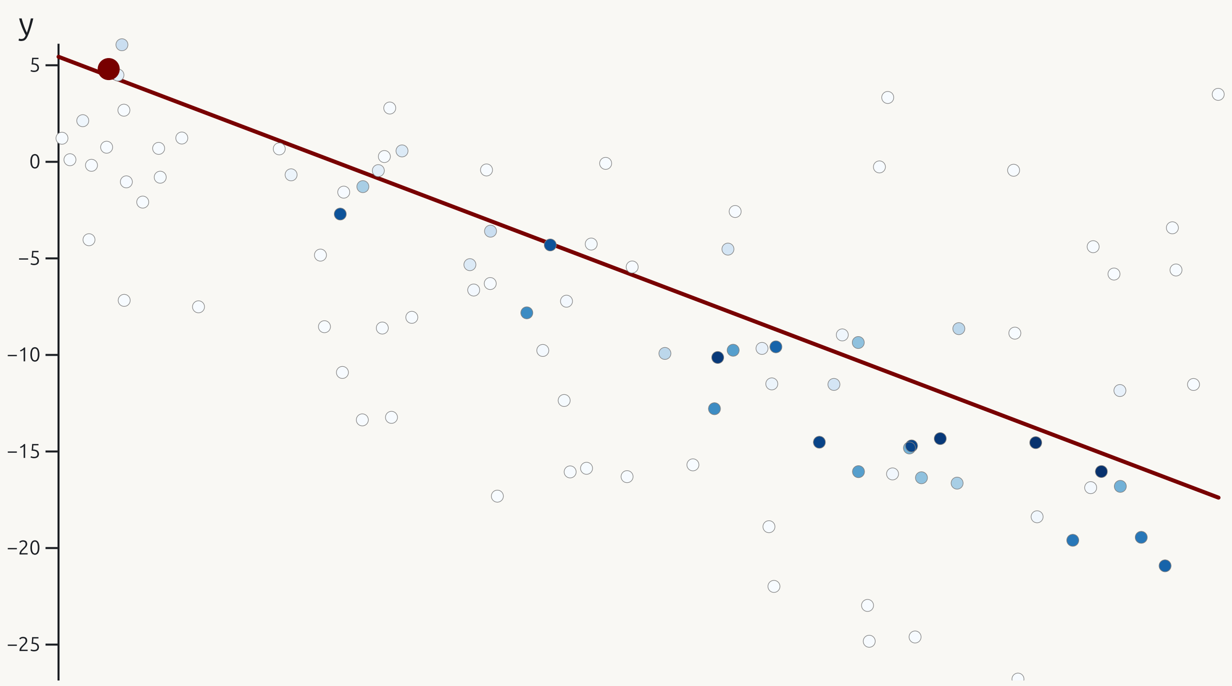
51

Simulate new data

Evolution: β_0 = 5.439 and β_1 = -11.415



Darker are closer to t = 51 (red dot)



Taxonomy of state-space models

- **Linear Gaussian**

$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{C}_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \boldsymbol{A}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-linear Gaussian** (additive) models

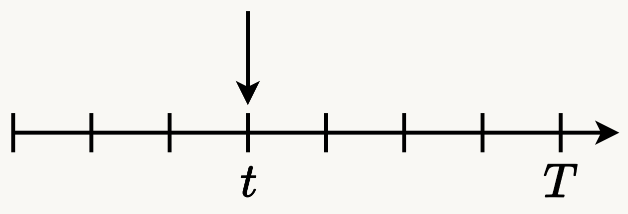
$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{c}_t(\boldsymbol{\theta}_t) + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \boldsymbol{a}_t(\boldsymbol{\theta}_{t-1}) + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\overset{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-Gaussian** - general distributions

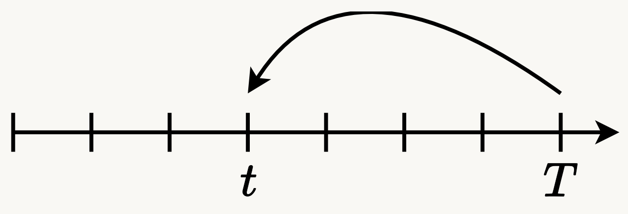
$$\begin{aligned} \boldsymbol{y}_t &\sim p_t(\boldsymbol{y}_t | \boldsymbol{\theta}_t) \\ \boldsymbol{\theta}_t &\sim q_t(\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t-1}) \end{aligned}$$

Bayesian inference in state-space models

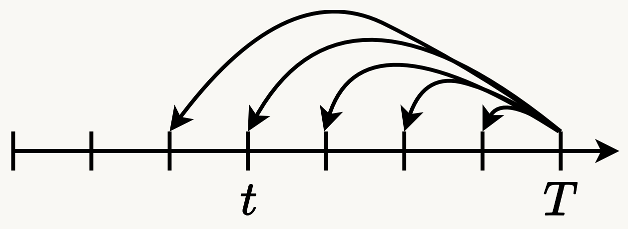
- **Filtering posterior** (instantaneous) $p(\boldsymbol{\theta}_t | y_{1:t})$



- **Smoothing posterior** (retrospective) $p(\boldsymbol{\theta}_t | y_{1:T})$



- **Joint smoothing posterior** $p(\boldsymbol{\theta}_{1:T} | y_{1:T})$



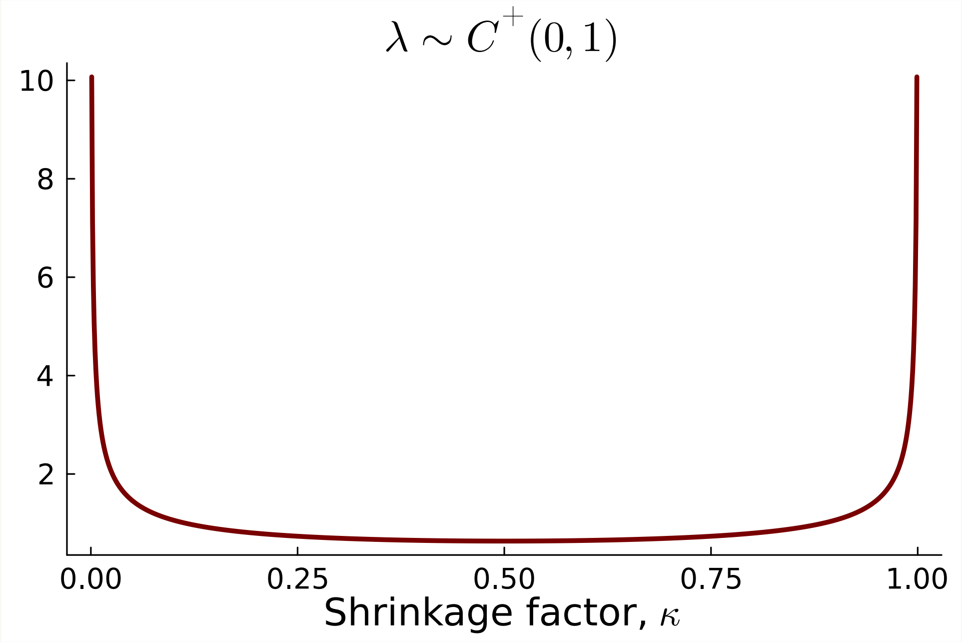
Global-local shrinkage priors - regression [1, 2]

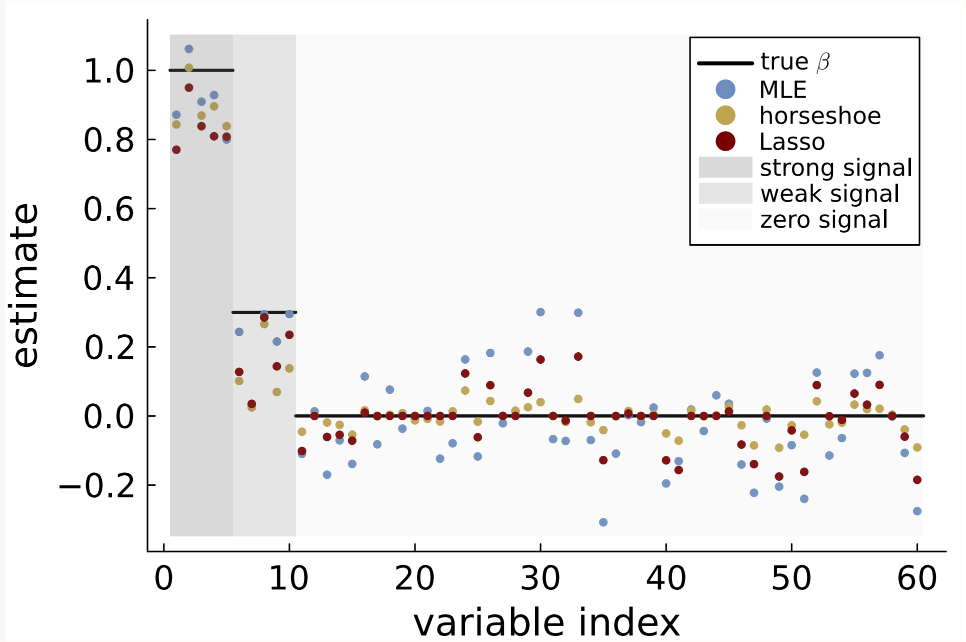
- Linear Gaussian regression with **horseshoe prior**

$$
\begin{aligned}
y_i &= \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \\
\beta_j &\overset{\text{ind}}{\sim} N(0, \tau^2 \lambda_j^2) \\
\tau &\sim C^+(0,1) \\
\lambda_j &\overset{\text{iid}}{\sim} C^+(0,1)
\end{aligned}
$$

- Posterior mean (for orthogonal $\boldsymbol{x}$) equals shrunken ML estimator

$$\tilde{\beta}_j = (1 - \kappa_j)\hat{\beta}_j$$





Dynamic global-local shrinkage priors for time series

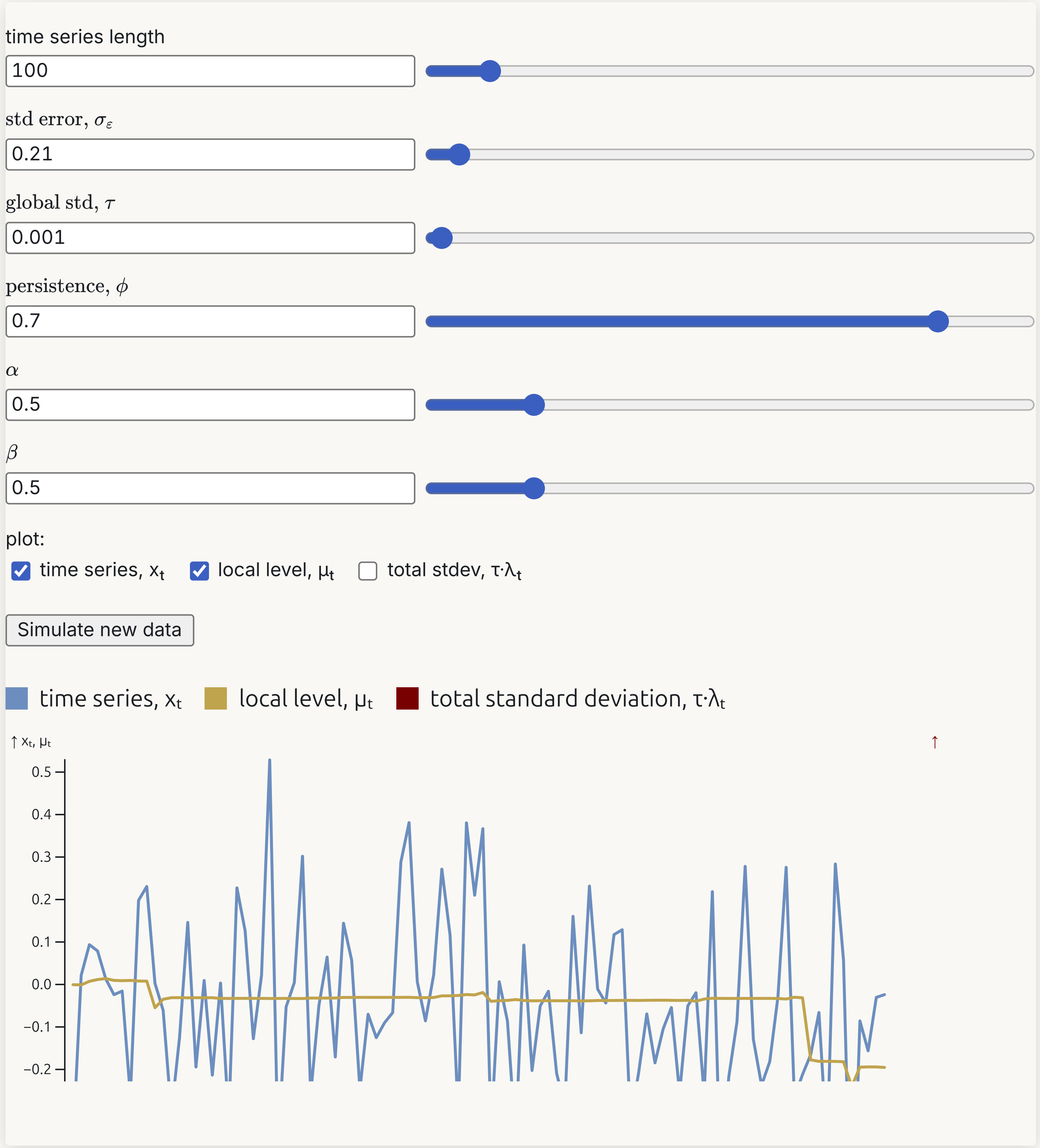
- **Local level model - dynamic shrinkage process (DSP) prior**

$$
\begin{aligned}
y_t &= \mu_t + \varepsilon_t & \varepsilon_t &\overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \\
\mu_t &= \mu_{t-1} + \nu_t, & \nu_t &\overset{\text{ind}}{\sim} N\big(0, \tau^2 \exp(h_t)\big) \\
h_t &= \rho h_{t-1} + \eta_t, & \eta_t &\overset{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1) \\
\tau &\sim C^+(0,1) & &
\end{aligned}
$$

- Global variance τ^2
- **Local variance $\lambda_t^2 = \exp(h_t)$**

- DSP becomes horseshoe when $\rho = 0$:

$$h_t \sim Z(\alpha, \beta, 0, 1) \iff \lambda_t = \exp(h_t/2) \sim C^+(0,1)$$



Logistic-Beta distribution (Fisher's Z)

- $X \sim \text{Beta}(\alpha, \beta)$ then

$$Z := \log\left(\frac{X}{1-X}\right) \sim Z(\alpha, \beta, 0, 1)$$

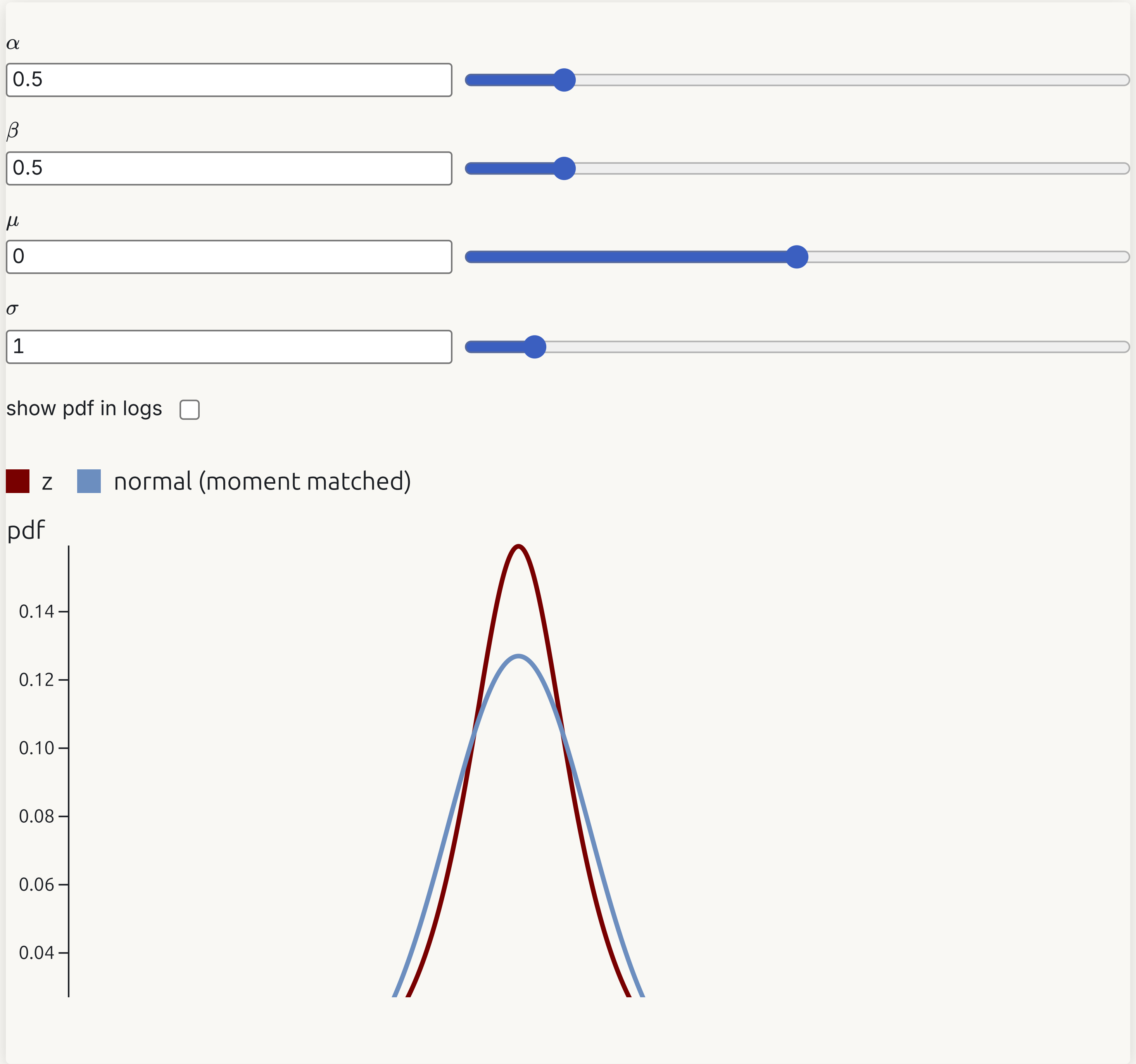
- Location-scale:

$$\mu + \sigma Z \sim Z(\alpha, \beta, \mu, \sigma)$$

- **Heavy tails** - log density decays linearly.

- Mean-variance mixture of Gaussians [3]

$$Z|\xi \sim N\left(\xi^{-1}(\alpha-\beta)/2, \xi^{-1}\right)$$
$$\xi \sim \text{PolyaGamma}(\alpha+\beta, 1)$$



Multi-seasonal ARMA with DSP prior [4]

- Seasonal AR with DSP evolution

$$\phi_{p,t}(L)\Phi_{P,t}(L^s)(y_t - \mu_t) = \varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} N(0, \sigma_t^2)$$

- SAR(1,1) as a **nonlinear Gaussian regression**

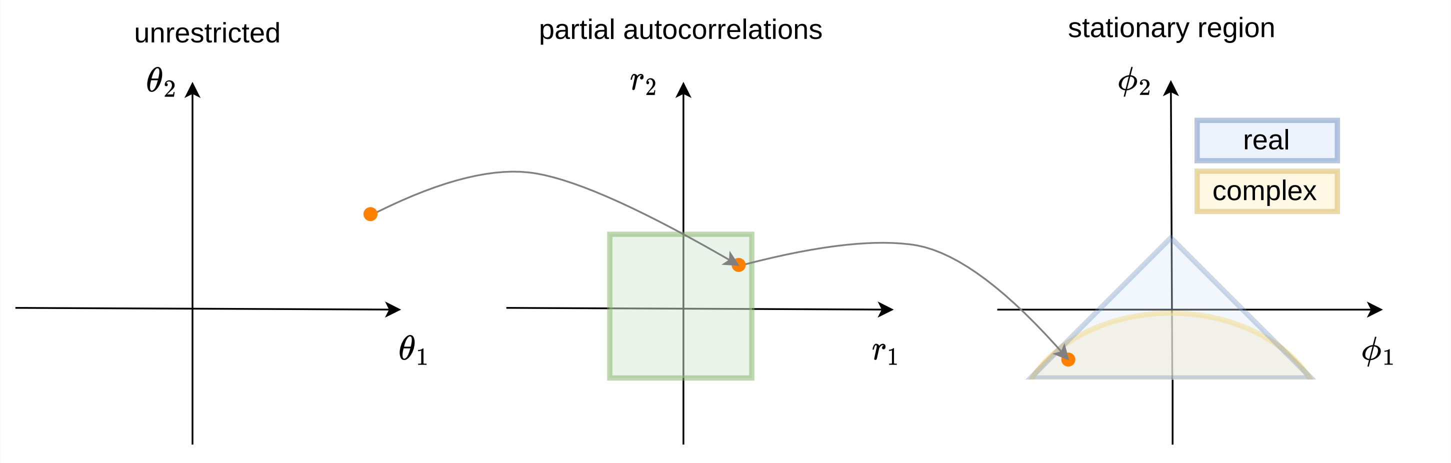
$$y_t = \boldsymbol{z}_t^\top \tilde{\boldsymbol{\phi}} + \varepsilon_t$$

$\tilde{\boldsymbol{\phi}} = (\phi_1, \Phi_1, -\phi_1\Phi_1)^\top$ and $\boldsymbol{z}_t = (x_{t-1}, x_{t-s}, x_{t-(1+s)})^\top$

- Extensions
  - [x] **Multi-seasonal** (daily, weekly, yearly cycle)
  - [x] ARMA and Exact likelihood
  - [x] Stochastic volatility with DSP priors
  - [ ] Multi-seasonal forecasting
  - [ ] Multi-seasonal VAR

Stability restrictions

Transformation to stability at every t



Lemma - uniform prior over the stability region [4, 5]

The AR parameters $\boldsymbol{\phi}$ are uniformly distributed over the stability region if, independently,

$$\theta_k \sim \begin{cases} t\left(k+1, 0, \frac{1}{\sqrt{k+1}}\right) & \text{for } k \text{ is odd} \\ t_{\text{skew}}\left(\frac{k}{2}, \frac{k+2}{2}, 0, \frac{1}{\sqrt{k+1}}\right) & \text{for } k \text{ is even} \end{cases}$$

- Transformation to partial autocorr matters
  - **Monahan**

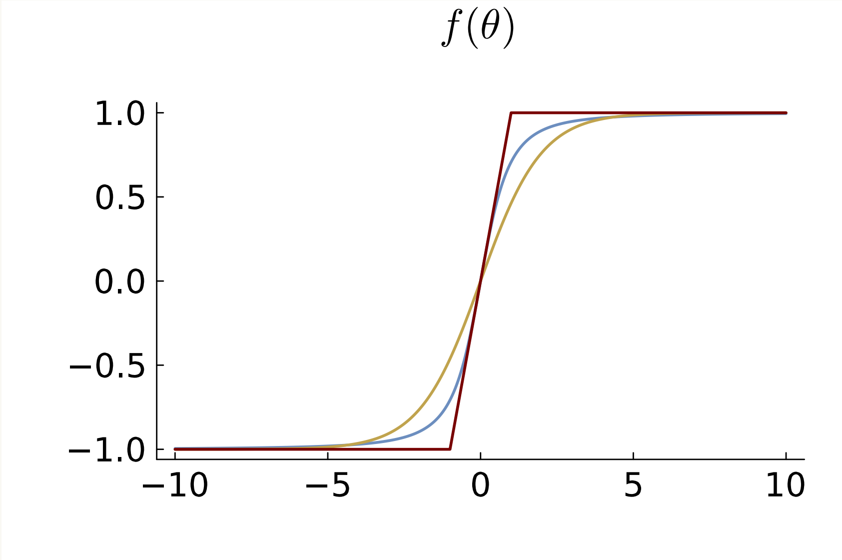
$$r = \frac{\theta}{\sqrt{1+\theta^2}}$$

  - **Tanh**

$$r = \frac{e^{\theta} - e^{-\theta}}{e^{\theta} + e^{-\theta}}$$

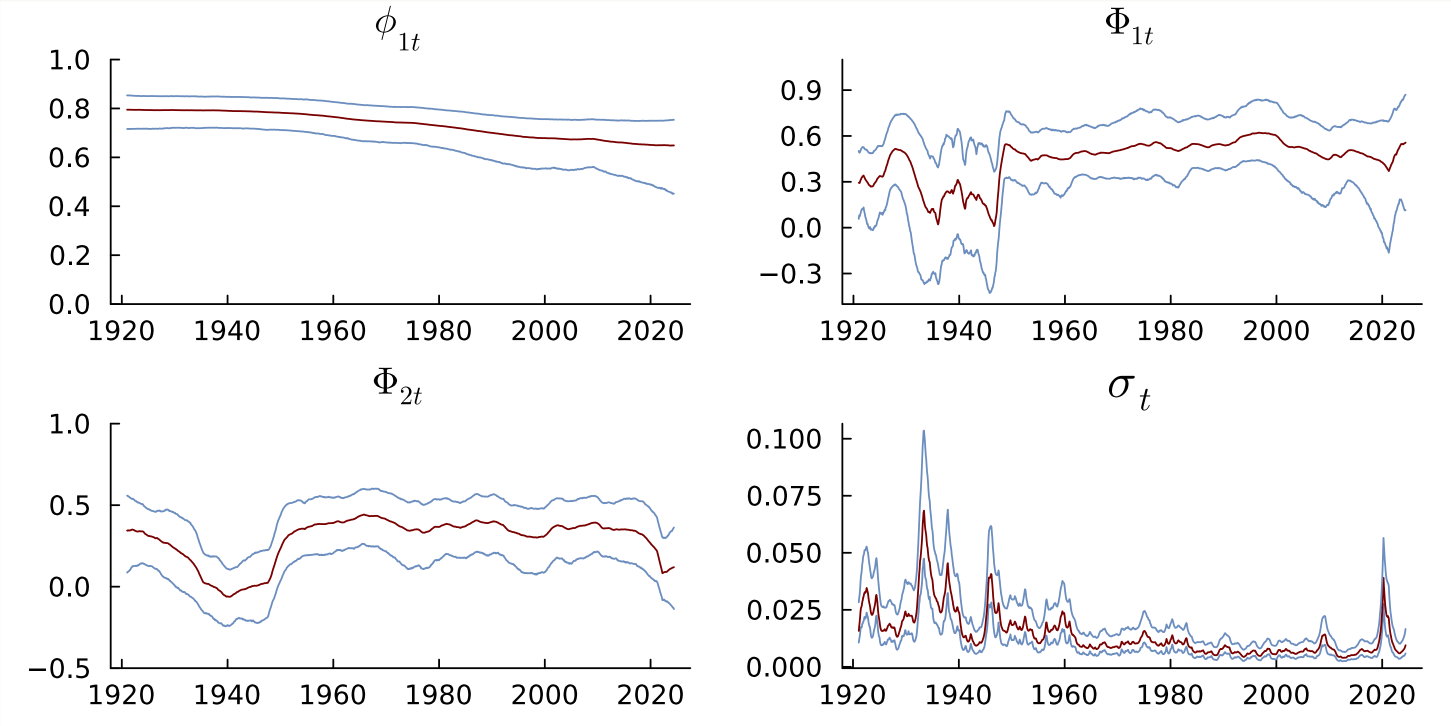
  - **Hard-tanh**

$$r = \begin{cases} -(1-\epsilon) & \text{if } \theta \leq -(1-\epsilon) \\ \theta & \text{if } -(1-\epsilon) < \theta < 1-\epsilon \\ 1-\epsilon & \text{if } \geq 1-\epsilon \end{cases}$$

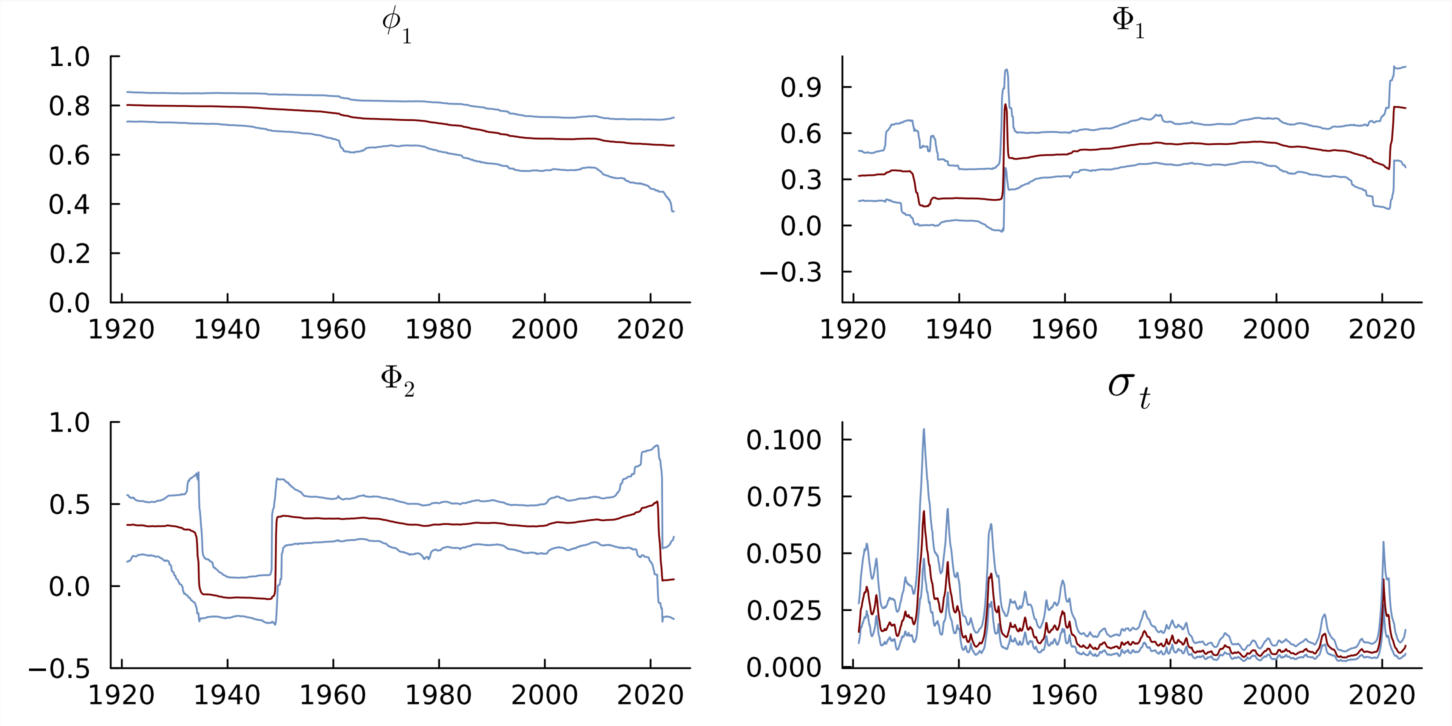


Seasonal AR with DSP prior - US industrial production 1919-2024

SAR(1,2) - Homoscedastic Gaussian



SAR(1,2) - DSP



Time-varying generalized linear models with DSP evolution

- **GLMs** with time-varying parameters

$$y_t | \boldsymbol{x}_t \sim \mathrm{ExpFamily}\left(k^{-1}(\boldsymbol{x}_t^\top \boldsymbol{\beta}_t), \varphi_t\right)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t, \qquad \boldsymbol{\nu}_t \overset{\text{iid}}{\sim} N\left(\mathbf{0}, \mathrm{Diag}(\exp(\boldsymbol{\tau}^2 \odot \boldsymbol{h}_t))\right)$$

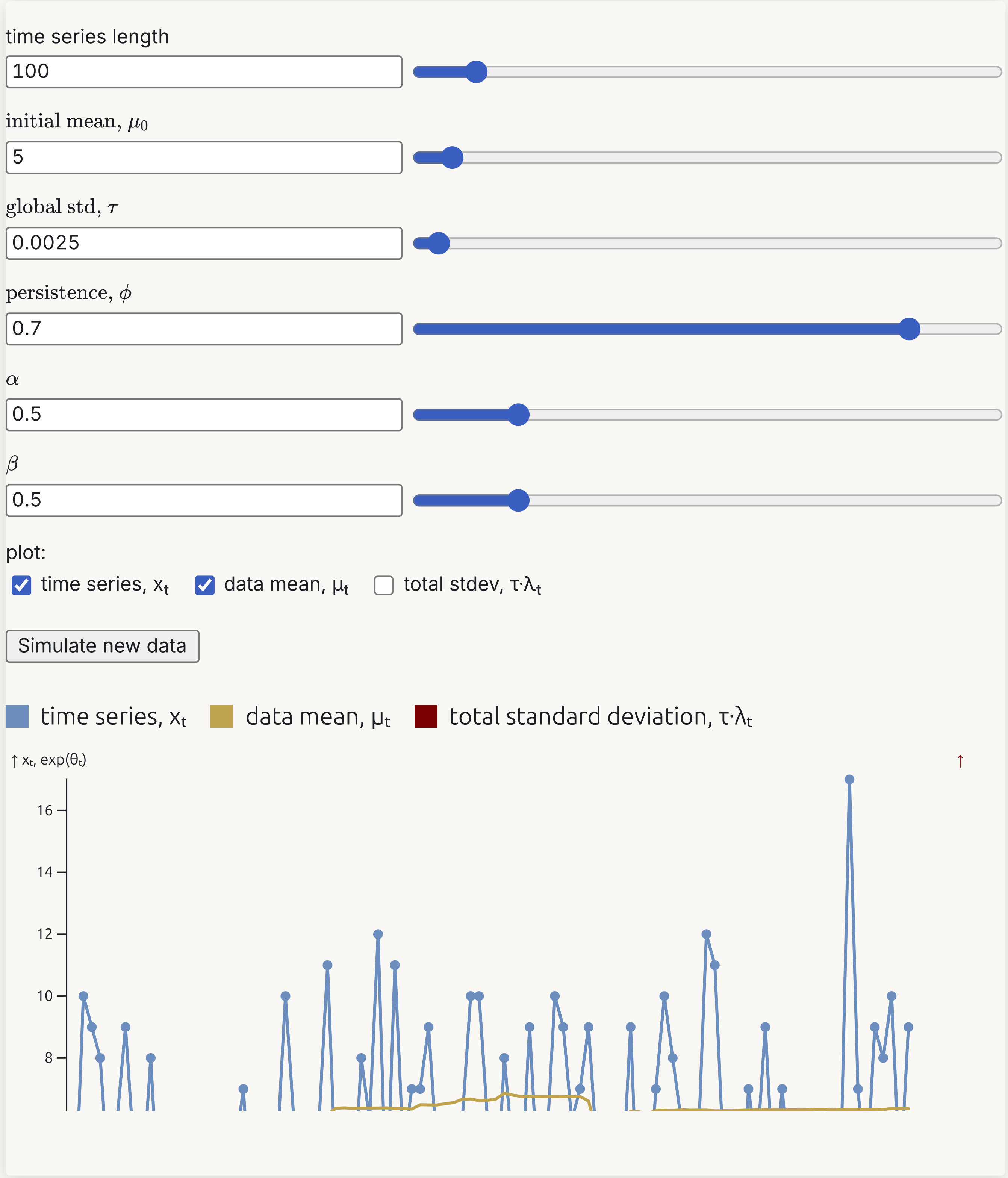
$$\boldsymbol{h}_t = \Phi \boldsymbol{h}_{t-1} + \boldsymbol{\eta}_t, \qquad \eta_{tk} \overset{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

- **Poisson regression** with DSP prior

$$y_t | \boldsymbol{x}_t \sim \mathrm{Pois}\left(\mu_t = \exp(\boldsymbol{x}_t^\top \boldsymbol{\beta}_t)\right)$$

- **Beta regression** with DSP prior

$$y_t | \boldsymbol{x}_t, z_t \sim \mathrm{Beta}\left(\mu_t = \mathrm{logistic}(\boldsymbol{x}_t^\top \boldsymbol{\beta}_t), \ \psi_t = \exp(\boldsymbol{z}_t^\top \gamma_t)\right)$$



Bayesian inference in state-space models with DSP prior

- Joint smoothing posterior ($\boldsymbol{\psi}$ holds all static parameters):

$$p(\boldsymbol{\theta}_{1:T}, \boldsymbol{h}_{1:T}, \boldsymbol{\psi} | y_{1:T})$$

- Gibbs sampling/MCMC

- Particle MCMC

- Hamiltonian Monte Carlo (HMC)

- Variational approximations

- Integrated Nested Laplace Approximation (INLA)?

Gibbs sampling in linear Gaussian model with DSP prior

- **Polya-Gamma augmentation** to turn Z-distribution into conditionally Gaussian. [6, 7]
- **Square-and-log trick** to turn variance $\exp(\boldsymbol{h}_t)$ into a mean $\boldsymbol{h}_t$. [8]
- Fast sampling of $\boldsymbol{h}_{1:T}$ using **sparse multivariate Gaussian** [9].
- **Forward-filtering-backward sampling** (FFBS) to sample $\boldsymbol{\theta}_{1:T}$ [10, 11]

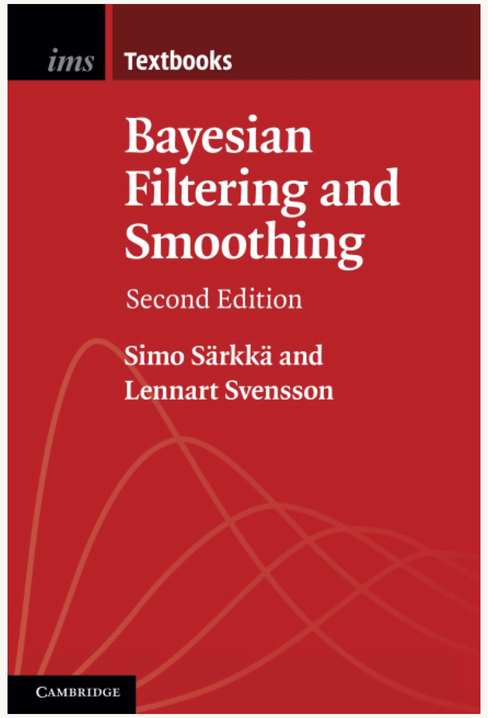
Gaussian approximation filters

- **Forward-filtering-backward sampling** for linear Gaussian models [10, 11]
  - Run Kalman filter $p(\boldsymbol{\theta}_t | \boldsymbol{y}_{1:t})$ forward for $t = 1, 2, \ldots, T$
  - Sample $p(\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t+1}, \boldsymbol{y}_{1:T})$ backward for $t = T, T-1, \ldots, 1$
- Non-linear/non-Gaussian: Kalman filter and FFBS not applicable.
- General idea: replace the filtering step with an approximate filter.
- **Prior propagation**: approximate the joint distribution

$$\begin{pmatrix} \boldsymbol{\theta}_t \\ \boldsymbol{\theta}_{t-1} \end{pmatrix} \mid \boldsymbol{y}_{1:t-1} \overset{\text{approx}}{\sim} N\left[\begin{pmatrix} \mathbb{E}(\boldsymbol{\theta}_t) \\ \mathbb{E}(\boldsymbol{\theta}_{t-1}) \end{pmatrix}, \begin{pmatrix} \mathbb{V}(\boldsymbol{\theta}_t) & \mathbb{C}(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t-1}) \\ \mathbb{C}(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t-1})^\top & \mathbb{V}(\boldsymbol{\theta}_{t-1}) \end{pmatrix}\right]$$

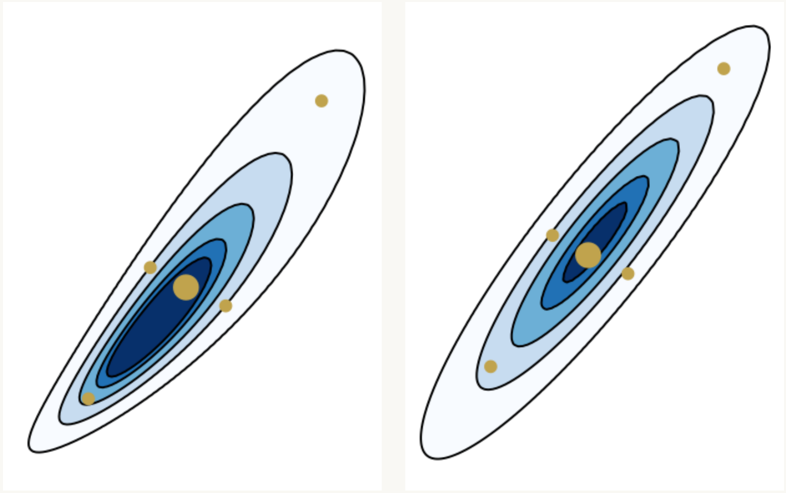
- **Observation update**: approximate the joint distribution

$$\begin{pmatrix} \boldsymbol{y}_t \\ \boldsymbol{\theta}_t \end{pmatrix} \mid \boldsymbol{y}_{1:t-1} \overset{\text{approx}}{\sim} N\left[\begin{pmatrix} \mathbb{E}(\boldsymbol{y}_t) \\ \mathbb{E}(\boldsymbol{\theta}_t) \end{pmatrix}, \begin{pmatrix} \mathbb{V}(\boldsymbol{y}_t) & \mathbb{C}(\boldsymbol{y}_t, \boldsymbol{\theta}_t) \\ \mathbb{C}(\boldsymbol{y}_t, \boldsymbol{\theta}_t)^\top & \mathbb{V}(\boldsymbol{\theta}_t) \end{pmatrix}\right]$$

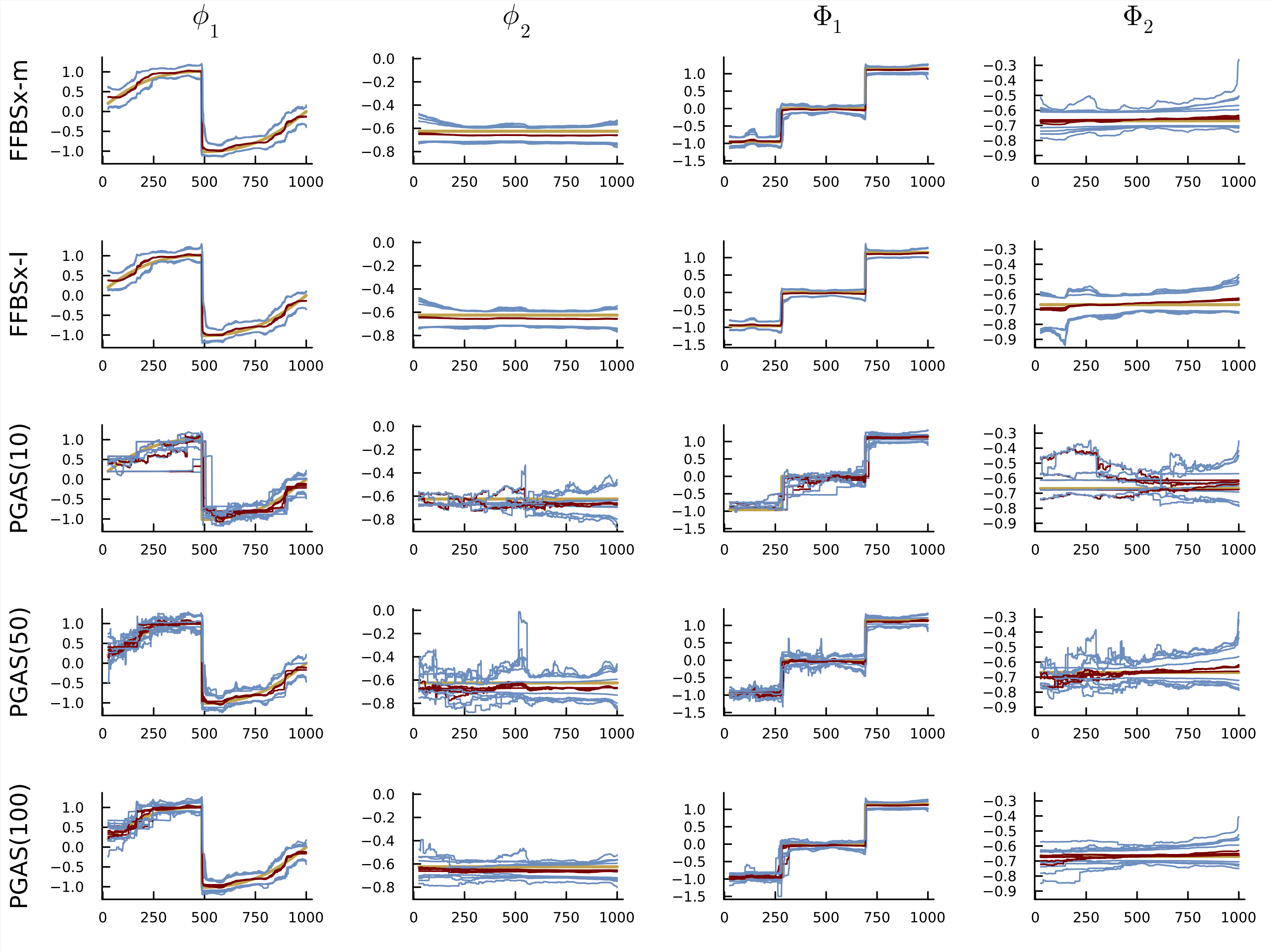


Gaussian approximation filters

- **Extended Kalman filter**
  - Linearize measurement/transition models
  - Apply standard Kalman filter
- **Unscented Kalman filter**
  - Propagate sigma point approx of prior/posterior
- **Prior linearization filter**
  - Linearize by statistical linear regression



Simulated example SAR(2,2) - MCMC convergence



Iterated Gaussian approximation filters

- Nonlinear Gaussian models
  - **Iterated extended Kalman filter** [12]
  - **Iterated unscented Kalman filter** [13]
  - **Posterior linearization filter** [14]
- Non-Gaussian models
  - **Posterior linearization filter - conditional moments version** [15]
    - Statistical linearization using $\mathbb{E}(Y|\boldsymbol{x})$ and $\mathbb{V}(Y|\boldsymbol{x})$.
  - **Laplace filter** [16, 17]
    - Direct Gaussian approximation of filtering posterior at every t

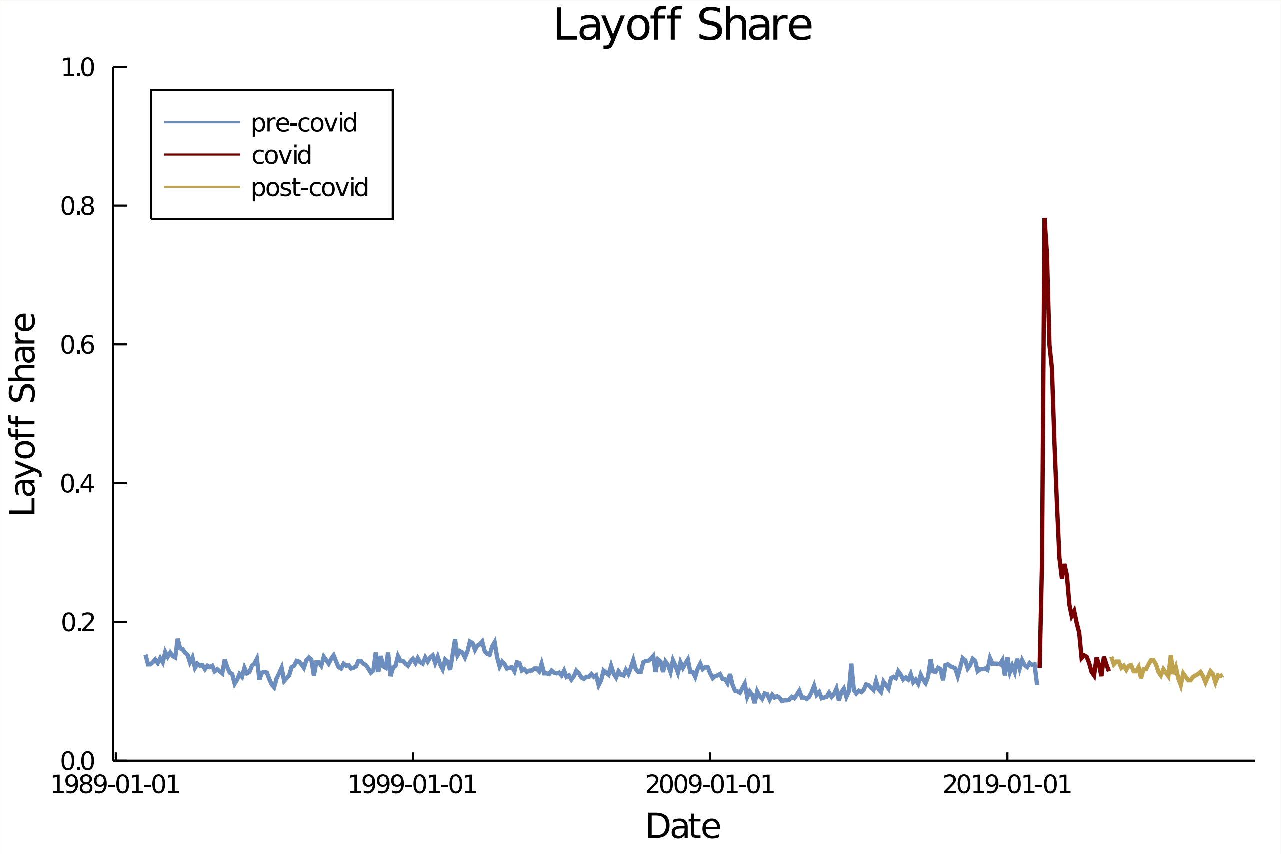
$$\boldsymbol{\theta}_t|y_{1:t} \overset{\text{approx}}{\sim} N(\tilde{\boldsymbol{\theta}}_{t|t}, \tilde{\boldsymbol{\Omega}}_{t|t})$$

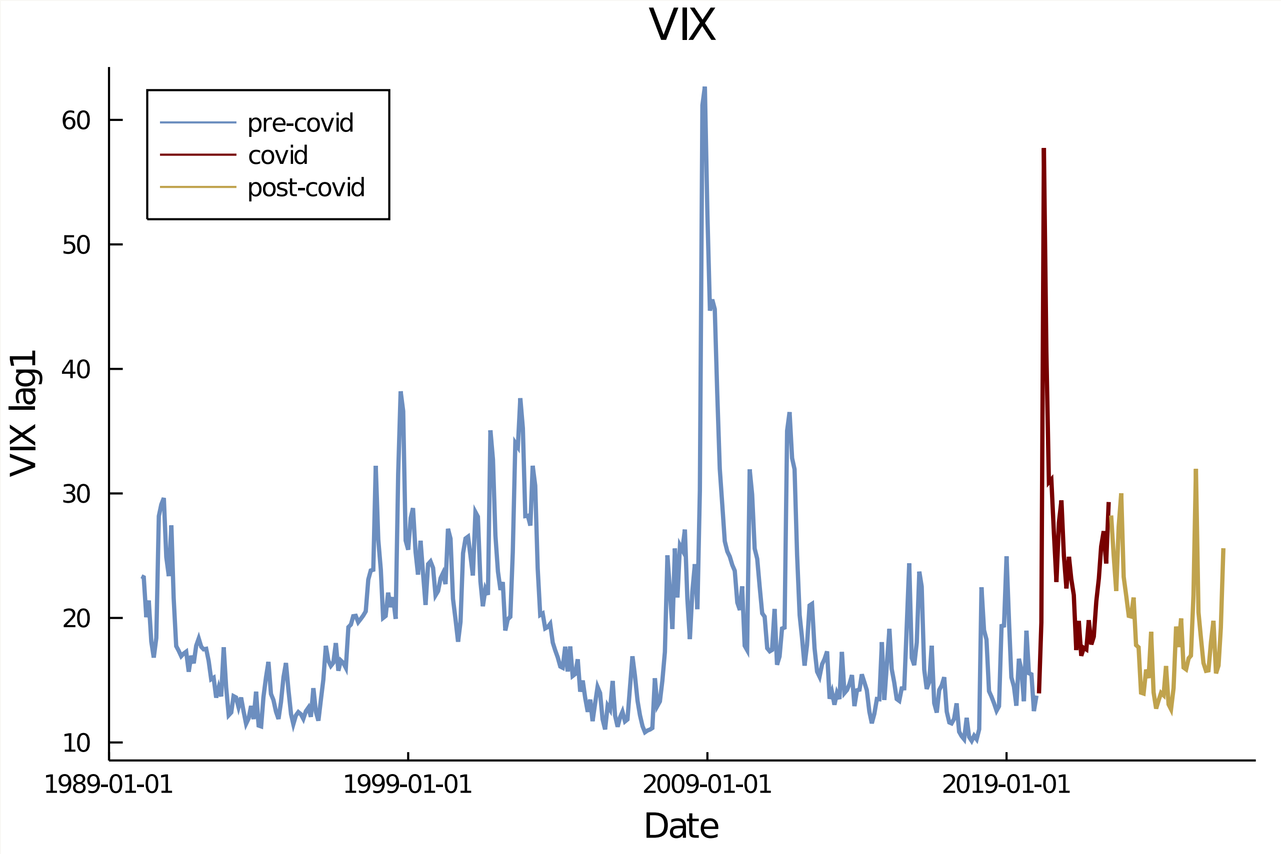
where $\tilde{\boldsymbol{\theta}}_{t|t}$ is posterior mode and $\tilde{\boldsymbol{\Omega}}_{t|t}$ is inverse of observed information.

    - **Automatic differentiation** makes it automatic.

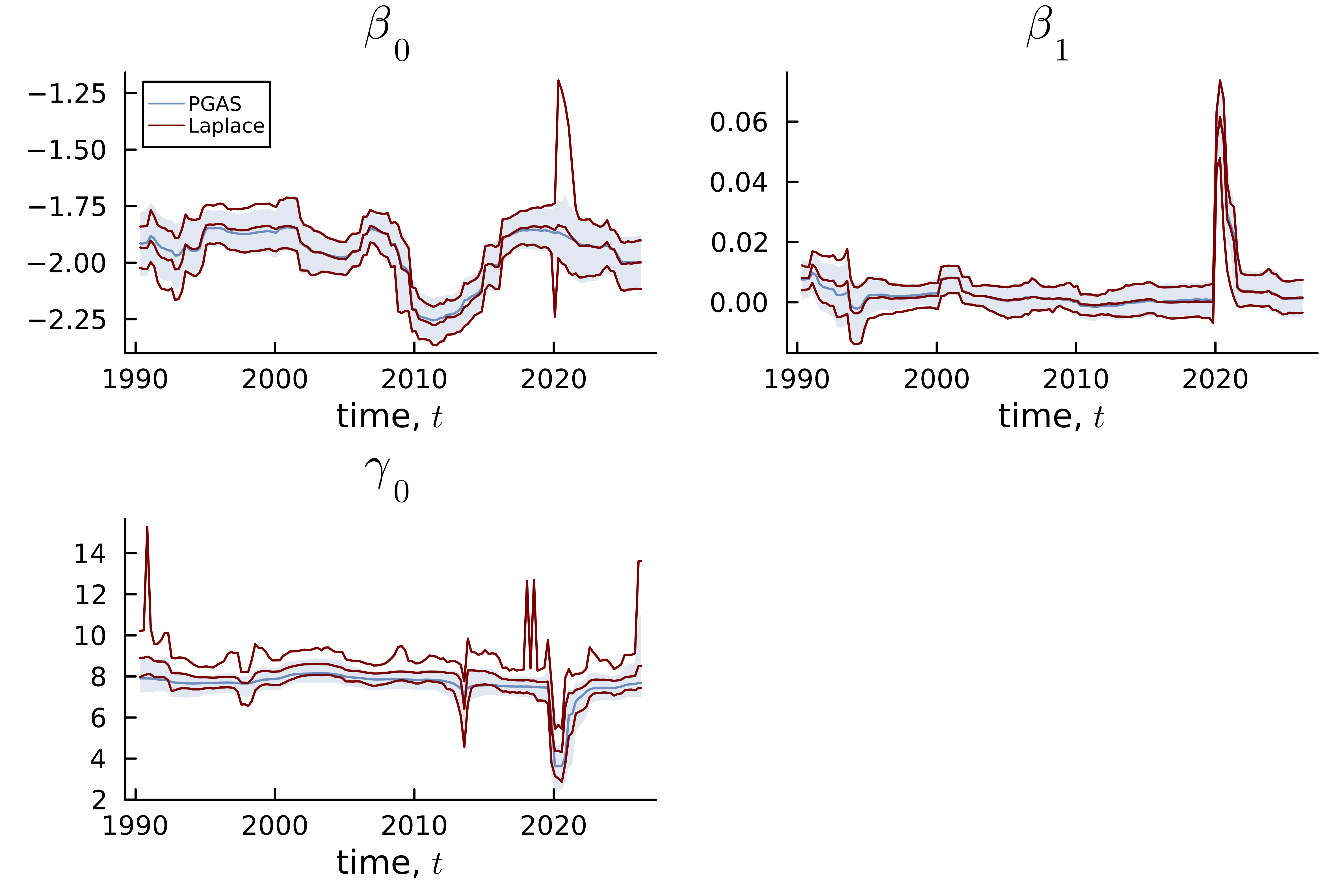
Time-varying Beta regression for US layoff proportion

- Monthly US data from 1990:1-2026:4 on:
  - `layoff` - job losers on layoff as % of total unemployed
  - `vix` - CBOE Volatility Index ("fear index")

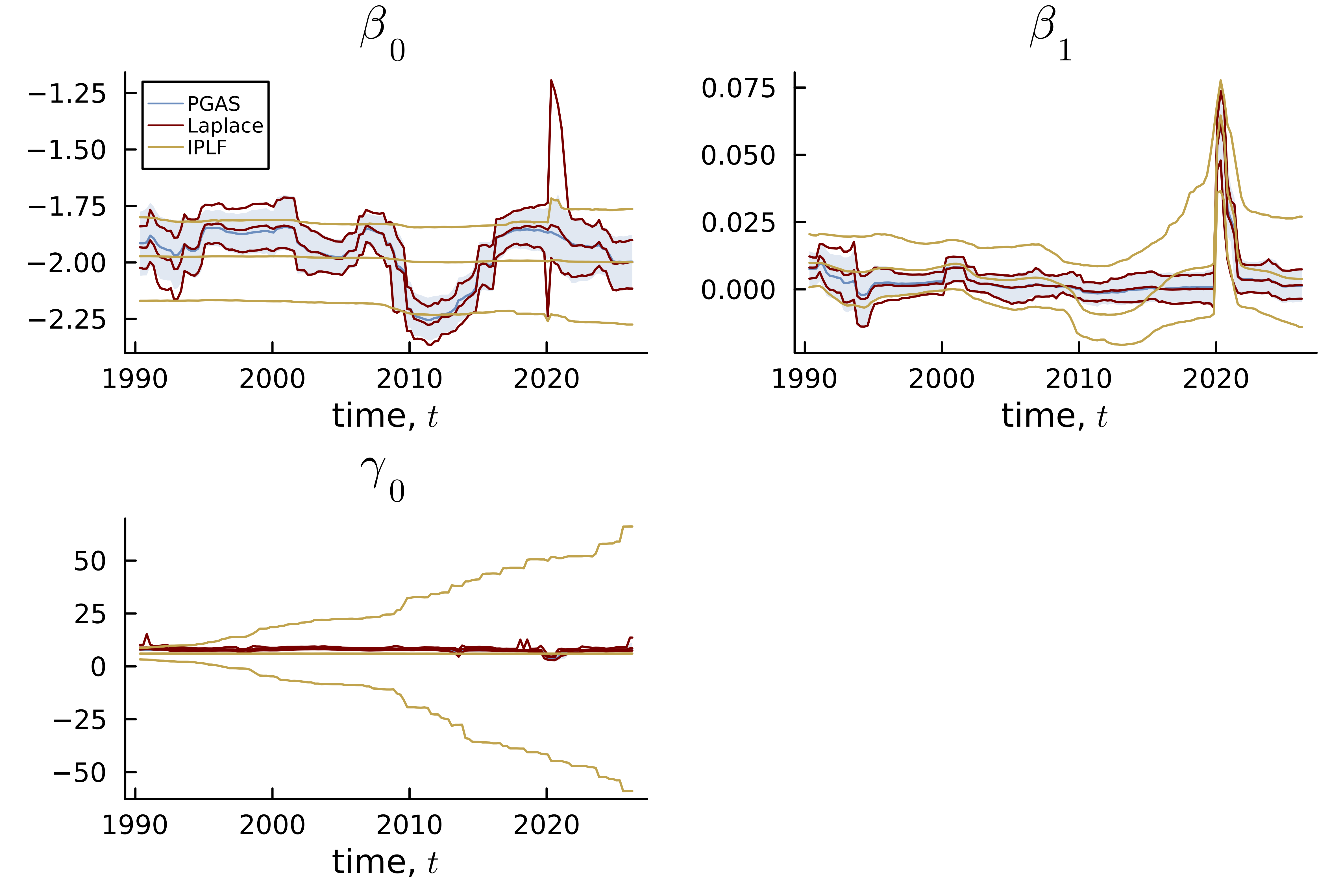




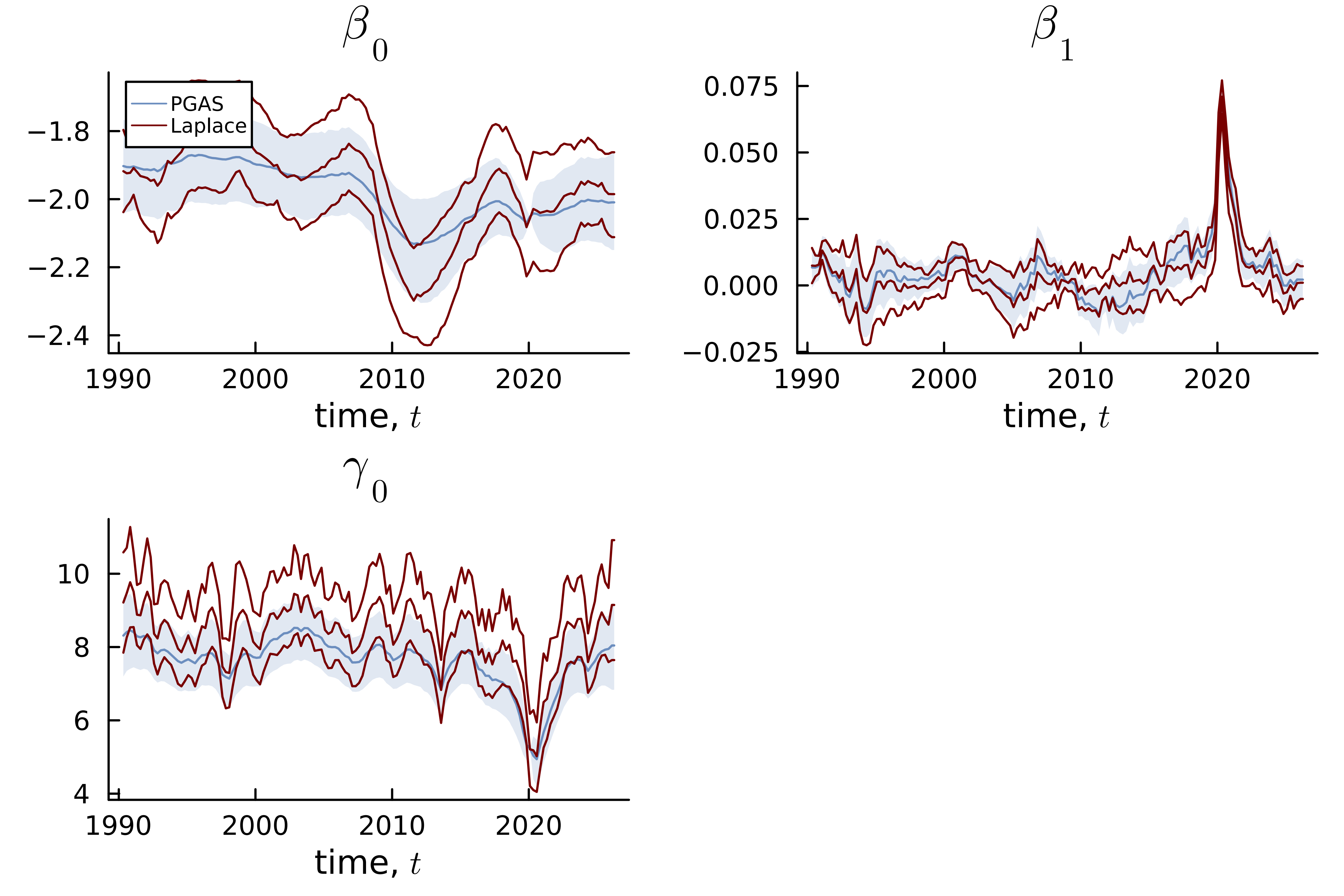
Beta regression for US layoff proportion - DSP prior



Linear approximations struggle with Beta regression

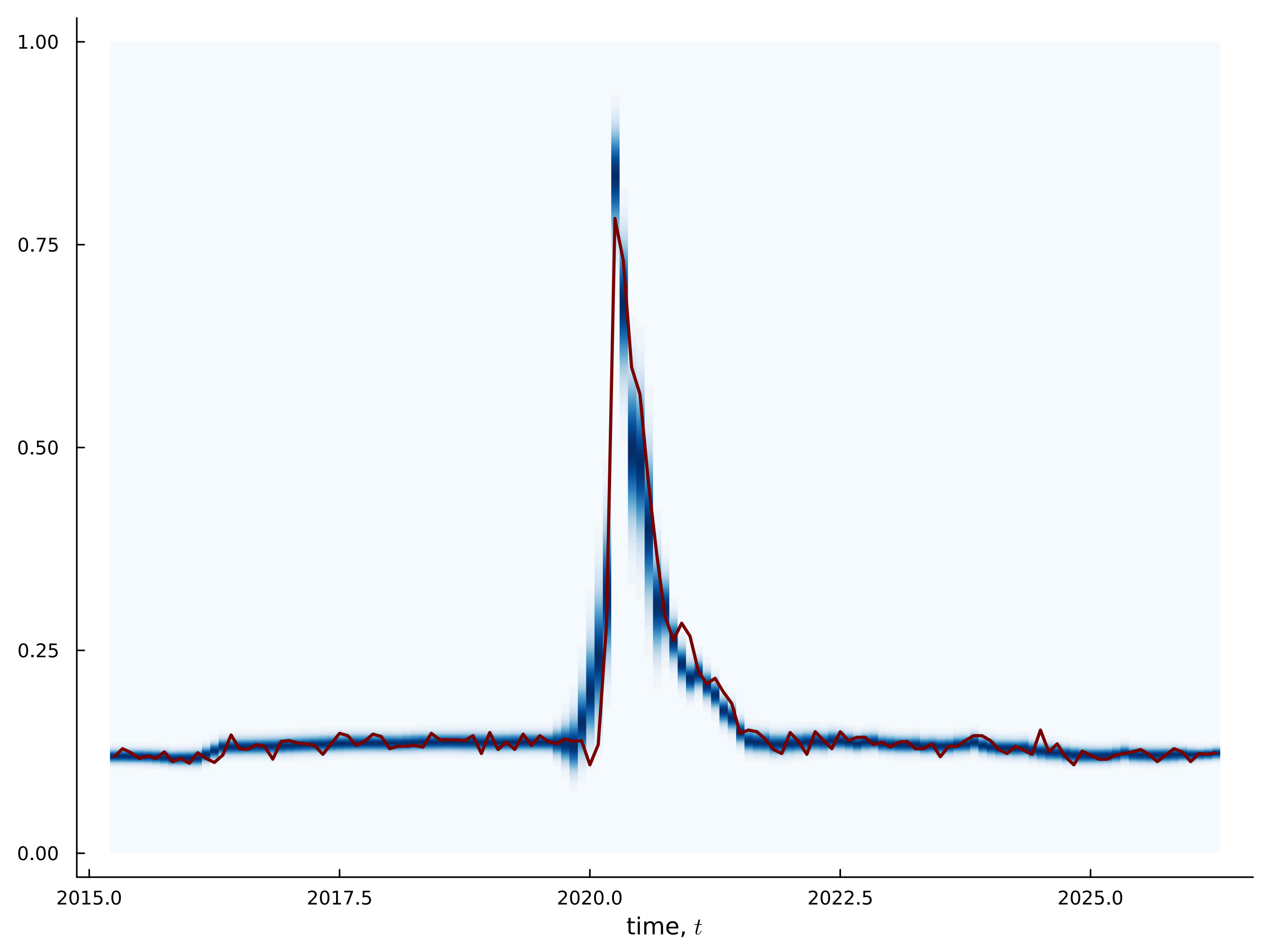


Beta regression for US layoff proportion - homoscedastic Gauss prior

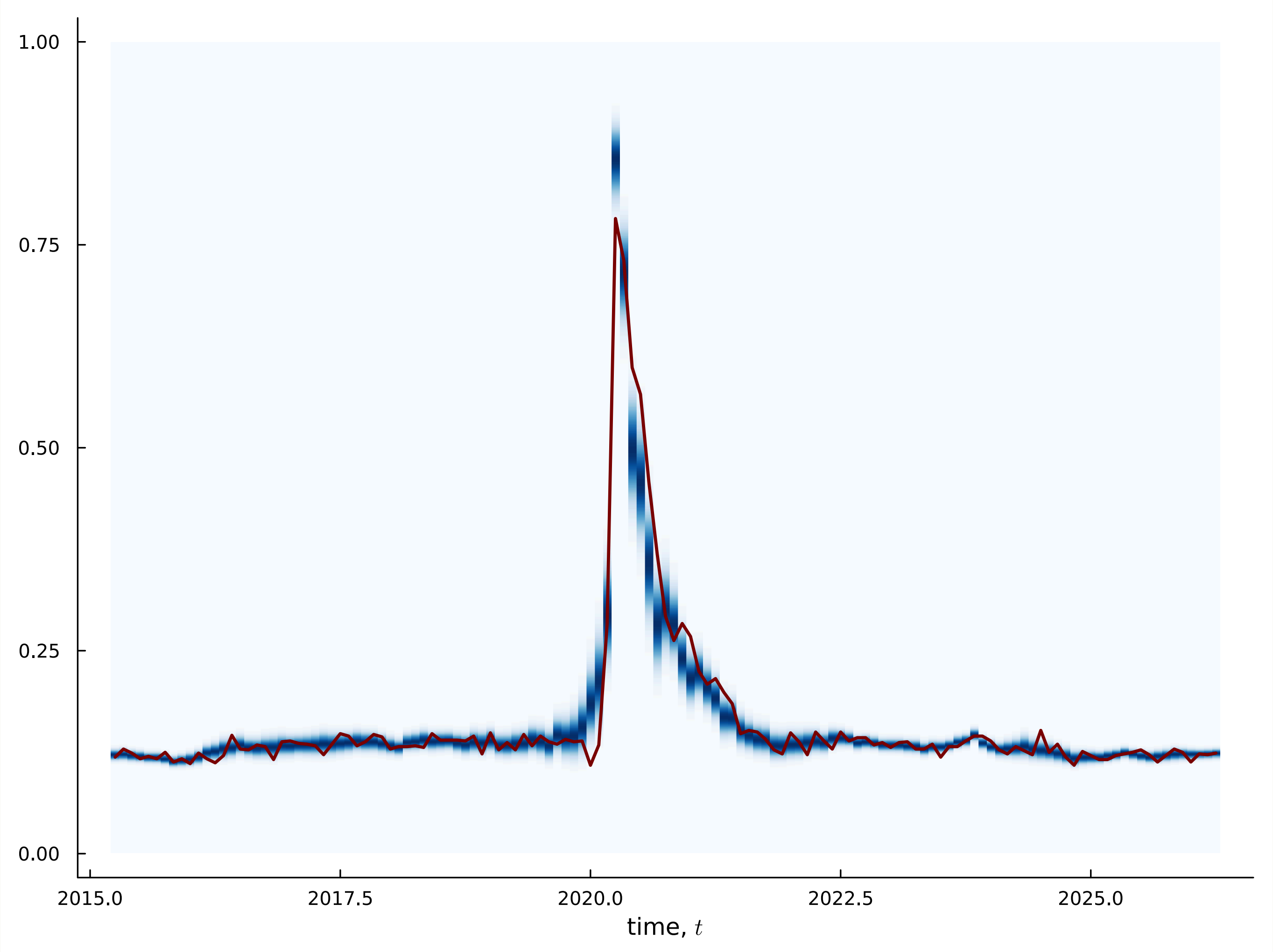


Beta regression for US layoff - fit at posterior median around pandemic

DSP



Homoscedastic Gaussian



Grouping observations for speed and Gaussianity

- Gaussian approximations at every time t.
- How Gaussian are the **likelihood contributions** $p(y_t|\boldsymbol{\theta}_t)$ in $\boldsymbol{\theta}_t$?
- No asymptotic Bernstein-von Mises effect to help.
- A tight propagated prior $\boldsymbol{\theta}_t \sim N(\hat{\boldsymbol{\theta}}_{t|t-1}, \boldsymbol{\Omega}_{t|t-1})$ helps a lot.
- But DSP can give large innovations, so $\boldsymbol{\Omega}_{t|t-1}$ can be large.

"When all the hope is gone, there is no reason for pessimism".
Aki Kaurismäki

- **Group observations** in G groups of J_g observations in group g.
- **Same state $\boldsymbol{\theta}_g$ for observations within a group**.

$$y_{gj}|\boldsymbol{x}_{gj} \sim \text{ExpFamily}\left(k^{-1}(\boldsymbol{x}_{gj}^\top \boldsymbol{\beta}_g), \varphi_g\right)$$

$$\boldsymbol{\beta}_g = \boldsymbol{\beta}_{g-1} + \boldsymbol{\nu}_g, \qquad \boldsymbol{\nu}_g \overset{\text{iid}}{\sim} N\left(\mathbf{0}, \text{Diag}(\exp(\boldsymbol{\tau}^2 \odot \boldsymbol{h}_g))\right)$$

$$\boldsymbol{h}_g = \Phi \boldsymbol{h}_{g-1} + \boldsymbol{\eta}_g, \qquad \eta_g \overset{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

where $j = 1, \ldots, J_g$ for $g = 1, \ldots, G$.

Grouping observations

- Related ideas:
  - Improve Gaussianity in Subsampling MCMC/HMC [18]
  - Improve linearity/Gaussianity in filtering [19]
  - Modeling moderate time evolutions [20]

- Two views:
  - interpolation for algorithmic performance
  - genuine model change: different time scales for observations and parameters.

- **Parameter evolution is automatically adapted** to different group sizes.
- **Massive speed-up**, particularly for sub-daily data.
- Example: 30 min electricity price. Group $48 \times 30 = 1440$ observations by month.
- Large groups: Sequential filtering of individual observations within group.

Natural parameter innovations

- Parameter evolution should adapt to:
  - Scales of covariates
  - Link functions
  - Model geometry
- **Natural gradients** [21, 22] in optimization. Fisher metric.
- Parameter evolution with **natural innovations**

$$y_t|\boldsymbol{x}_t \sim \mathrm{ExpFamily}\Big(k^{-1}(\boldsymbol{x}_t^\top \boldsymbol{\beta}_t), \varphi_t\Big)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{S}(\boldsymbol{\beta}_{t-1})\boldsymbol{\nu}_t, \qquad \boldsymbol{\nu}_t \overset{\text{iid}}{\sim} N\big(\mathbf{0}, \mathrm{Diag}(\exp(\boldsymbol{\tau}^2 \odot \boldsymbol{h}_t))\big)$$

$$\boldsymbol{h}_t = \boldsymbol{\Phi}\boldsymbol{h}_{t-1} + \boldsymbol{\eta}_t, \qquad \eta_{tk} \overset{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

where $\boldsymbol{S}(\boldsymbol{\beta}_{t-1}) = \bar{I}^{-1/2}(\boldsymbol{\beta}_{t-1})$

- $\bar{I}(\boldsymbol{\beta}) = \frac{1}{T} I(\boldsymbol{\beta})$ is the **Fisher information** from an “average” observation.
- Non-linear transition model. Prior propagation step: $S(\boldsymbol{\beta}_{t-1}) \approx S(\hat{\boldsymbol{\beta}}_{t-1|t-1})$.

Ongoing and future work

- **Multivariate** extensions.
- **Forecasting**.
- More **efficient sampling of DSP**.
- Better **prior elicitation for DSP**.
- **Metropolis-Hastings correction** by blocking the state. Filtering version of [16].
- **Delayed rejection**:
  - first try with cheap proposal (e.g. EKF)
  - if rejected try more expensive proposal (e.g. Iterated EKF with line search)

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